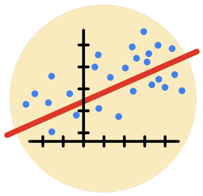
Course Four

Regression Analysis: Simplifying Complex Data Relationships



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. As a reminder, this document is a resource that you can reference in the future, and a guide to help you consider responses and reflections posed at various points throughout projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- [ ] Complete the questions in the Course 4 PACE strategy document
- [ ] Answer the questions in the Jupyter notebook project file
- [ ] Build a multiple linear regression model
- [ ] Evaluate the model
- [ ] Create an executive summary for team members

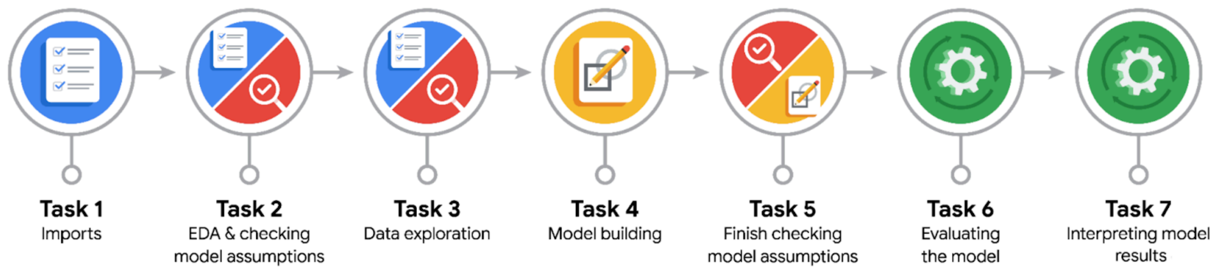
Relevant Interview Questions

Completing the end-of-course project will empower you to respond to the following interview topics:

- Describe the steps you would take to run a regression-based analysis
- List and describe the critical assumptions of linear regression
- What is the primary difference between R^2 and adjusted R^2?
- How do you interpret a Q-Q plot in a linear regression model?
- What is the bias-variance tradeoff? How does it relate to building a multiple linear regression model? Consider variable selection and adjusted R^2.

Reference Guide

This project has seven tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- Who are your external stakeholders for this project?

Partes Interesadas Externas (External Stakeholders)

- **Maika Abadi (COO):** Needs to understand video feature correlations with verified status to align platform business operations. / Necesita entender la correlación de las métricas con la verificación para alinear las operaciones comerciales.
- **Mary Joanna Rodgers (Executive Lead):** Final recipient of the executive summary to evaluate milestones and approve resources. / Destinataria final del resumen ejecutivo para evaluar hitos y autorizar recursos.
- **Trust and Safety Team:** Primary end-users; requires baseline insights to optimize automated content moderation pipelines. / Usuarios finales; requieren los insights base para optimizar la moderación automatizada.

- What are you trying to solve or accomplish?

- *N:* Predict verified_status using video performance metrics to provide a foundational stepping stone for the final claim vs. opinion classifier. This will minimize false positives in automated moderation and protect legitimate creators.

- *ES:* Predecir verified_status mediante métricas de rendimiento de videos como base para el clasificador final de reclamos vs. opiniones. Esto minimizará los falsos positivos en la moderación automatizada y protegerá a los creadores legítimos

- What are your initial observations when you explore the data?

Content Bias: Verified users overwhelmingly post opinions rather than claims. / Los usuarios verificados publican abrumadoramente opiniones en lugar de reclamos.

Right-Skewed Metrics: Engagement data is heavily right-skewed; extreme virality is rare and driven by unverified accounts. / Las métricas presentan un sesgo marcado a la derecha; la viralidad masiva es inusual y proviene de cuentas no verificadas.

Organic Outliers: Extreme values reflect organic platform virality, not data errors. / Los valores extremos son anomalías operativas legítimas que reflejan la viralidad de la plataforma, no errores.

Multicollinearity: Predictor variables move together in tight linear correlations, threatening model stability. / Las variables predictoras muestran una fuerte correlación lineal, amenazando la estabilidad del modelo.

- What resources do you find yourself using as you complete this stage?

1. Librerías del Ecosistema de Python (Stack Técnico)

- **EN:** Python ecosystem (`pandas`, `numpy` for data manipulation; `scikit-learn` for modeling; `seaborn`/`matplotlib` for EDA), the PACE framework, and internal TikTok project briefs.
- **ES:** Ecosistema de Python (`pandas`, `numpy` para datos; `scikit-learn` para modelado; `seaborn`/`matplotlib` para EDA), el marco PACE e informes internos de TikTok.

PACE: Analyze Stage

- What are some purposes of EDA before constructing a multiple linear regression model?

Assumptions Check: Verify linearity via scatter plots and residual normality via histograms or Q-Q plots. / Evaluar la linealidad con gráficos de dispersión y la normalidad de residuos con histogramas o gráficos Q-Q.

Multicollinearity: Generate correlation heatmaps to identify redundant features that destabilize model interpretability. / Generar mapas de calor para identificar variables redundantes que desestabilicen la interpretabilidad.

Outlier Assessment: Utilize box plots to detect extreme points that could disproportionately skew the regression line. / Utilizar diagramas de caja para detectar puntos extremos que puedan sesgar la línea de regresión.

Feature Engineering: Guide necessary mathematical transformations (e.g., log scales) and categorical encodings. / Guiar transformaciones matemáticas necesarias (ej. escalas logarítmicas) y la codificación de variables categóricas

- Do you have any ethical considerations at this stage?

- **Algorithmic Bias:** Over-relying on viral metrics risks automatically penalizing smaller, unverified creators when they achieve traction. / Depender en exceso de métricas virales corre el riesgo de penalizar automáticamente a creadores pequeños al volverse virales.

- **User Privacy:** Aggregating engagement data to dynamically evaluate account legitimacy must strictly adhere to global privacy regulations. / Agrupar datos de interacción para evaluar la legitimidad de una cuenta debe cumplir con las normativas de privacidad.
- **Explainability:** Maintaining transparency is critical so creators have a fair and clear automated moderation appeal process. / Preservar la transparencia es crítico para garantizar un proceso justo de apelación ante la moderación.
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PACE: Construct Stage

- Do you notice anything odd?

- **Model Contradiction:** Introductory text requests Multiple Linear Regression, but the binary outcome (verified_status) mathematically mandates Logistic Regression as specified in executive emails. / El texto introductorio pide Regresión Lineal Múltiple, pero el resultado binario exige Regresión Logística, tal como aclaran los correos.

- **Reverse Causality:** The model uses video metrics to predict verification status, but operationally, holding a verified badge is what drives baseline engagement curves. / El modelo usa métricas para predecir la verificación, cuando tener la insignia es lo que impulsa el engagement base.

- Can you improve it? Is there anything you would change about the model?

- **Log Transformations:** Apply $log(x+1) to skewed engagement features to stabilize calculated weights. / Aplicar $log(x+1) a las variables sesgadas para estabilizar los pesos calculados.
- **Engagement Ratios:** Drop raw absolute volumes and engineer rates (e.g., like_to_view_ratio) to eliminate structural multicollinearity. / Diseñar tasas (ej. like_to_view_ratio) para eliminar la multicolinealidad estructural.
- **Regularization:** Implement Lasso (L1) tuning to drive non-essential coefficients to zero and prevent overfitting. / Implementar regularización Lasso (L1) para reducir coeficientes no esenciales a cero y evitar el sobreajuste.
- **Architecture Upgrade:** Use this linear model strictly as a performance baseline and transition to tree-based ensembles (**Random Forest / XGBoost**). / Usar este modelo lineal como línea base y transicionar a ensambles basados en árboles (**Random Forest / XGBoost**).

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- What resources do you find yourself using as you complete this stage?

- **EN:** The same technical stack and project guidelines detailed in the Plan Stage (Page 6-7) were maintained.
- **ES:** Se mantuvo el mismo stack técnico y lineamientos detallados en la Etapa de Planificación (Páginas 6-7).

PACE: Execute Stage

- What key insights emerged from your model(s)?

- **Classification Limits / Límites de Clasificación:** Evaluated $7,154$ instances. High sensitivity (captured $3,099$ out of $3,605$ verified accounts), but massive Type I error (falsely flagged $1,960$ unverified accounts as verified).
- **Behavioral Patterns / Patrones de Variables:** Verification is driven by deep engagement (text_length and video_comment_count) rather than raw, volatile virality (video_view_count), which yielded negative coefficients.
- **Operational Risk / Riesgo Operativo:** Linear models cannot cleanly map TikTok's non-linear interaction patterns. Pipeline must scale to tree ensembles.

What business recommendations do you propose based on the models built?

Halt Automated Badging / Suspender Automatización: Do not deploy this model to automatically grant badges; its 55.2% False Positive rate would dilute platform trust.

Human-in-the-Loop Queue / Moderación Híbrida: Use predictions as a triage filter. Route predicted positives to a prioritized human verification queue to optimize manual auditing.

Incentive Shift / Cambio de Incentivos: Reward creators who foster sustainable, deep community dialogue (comments) over raw, volatile virality.

Technical Migration / Migración Técnica: Upgrade the engineering pipeline to **Random Forest** or **XGBoost** to enforce a tighter decision boundary and reduce false alarms.

- What potential recommendations would you make?

    - **EN:** Please refer directly to the **Strategic Business Recommendations** structured in the previous section (**Page 19-21**).
    - **ES:** Por favor, remitirse directamente a las **Recomendaciones Estratégicas de Negocio** estructuradas en la sección anterior (**Páginas 19-21**).

- Do you think your model could be improved? Why or why not? How?

- **Why / Por qué:** Yes. It suffers from a high False Positive rate (55.2%) because social media metrics naturally violate the linearity assumptions of Logistic Regression.
- **How / Cómo:** By migrating to non-linear tree ensembles (**Random Forest / XGBoost**), optimization of decision thresholds, and engineering interaction ratios.

- What business/organizational recommendations would you propose based on the models built?

- **EN:** Organizational deployment alignment follows the exact risk-mitigation guidelines structured in **Page 19-21**.
- **ES:** La alineación del despliegue organizacional sigue las mismas pautas de mitigación de riesgos estructuradas en las **Páginas 19-21**.

- Given what you know about the data and the models you were using, what other questions could you address for the team?

- **Virality Saturation / Saturación de Viralidad:** At what exact threshold do massive views stop signaling authority and start indicating bot farm activity?
- **Description Sentiment / Análisis de Sentimiento:** How does caption sentiment and NLP keyword patterns impact an account's ban risk?
- **Recovery Metrics / Métricas de Recuperación:** What interaction metrics allow a profile under review to safely return to an active standing?
- **False Positive Post-Mortem / Mapeo de Errores:** What feature combinations caused the linear model's $1,960$ False Positives, to build better rules for tree models?

- Do you have any ethical considerations at this stage?

Operational Equity / Equidad: A 55.2% error rate risks diluting platform integrity if unverified accounts are wrongfully promoted.

Algorithmic Exclusion / Exclusión: Over-indexing on description length and comment count can penalize niche or non-native language creators.

System Manipulation / Manipulación: Presenting these metrics might incentivize users to game the system using engagement pods instead of creating authentic content.

Human Safeguard / Supervisión Humana: Automated models must never hold absolute authority over a creator's status; a Human-in-the-Loop appeal workflow is mandatory.